Ejercicio Realizado por ADAIR HERNANDEZ VALDIVIA

Regla de aprendizaje de Rosenblatt

(4 puntos posibles)

El primer esfuerzo para que el perceptrón aprendiera a partir de datos fue la regla de aprendizaje presentada por Frank Rosenblatt. En este notebook aplicaremos dicha regla para observar el comportamiento sobre un conjunto de datos sintético.

Ejecuta/Corre

```
import numpy as np
import matplotlib as mpl
import matplotlib as mpl
import matplotlib.pyplot as plt

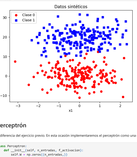
from sklearn.inspection import DecisionBoundaryDisplay
from matplotlib.colors import LinearSegmentedCmap

colors = ['red', 'blue']

# Generar datos sintéticos
X, y = mf.make(500,random)

cmap = LinearSegmentedColorMap.from_list('mycmap',
colors)

# Visualizar los datos
fig, ax = plt.subplots()
ax = fig.subplots()
ax.scatter(X_train[y == 1], X_train[y == 1], c=cmap(1), label='Clase 0')
ax.scatter(X_train[y == 0], X_train[y == 0], c=cmap(0), label='Clase 1')
ax.set_ylabel('$x_2$')
ax.set_xlabel('$x_1$')
ax.set_title('Datos sintéticos')
ax.legend(loc='upper left')
plt.show()
```



Perceptrón

A diferencia del ejercicio previo. En esta ocasión implementaremos el perceptrón como una clase

```
class Perceptron:
    def __init__(self, n_entradas, f_activacion):
        self.n_entradas = n_entradas
        self.b = 0
        self.f_activacion = f_activacion

    def inferencia(self, X):
        h = np.dot(X, self.w) + self.b
        return self.f_activacion(h)

def sigmoide(x):
    return np.exp(x) / (1 + np.exp(x))
```

Inicialización

El primer paso consiste en inicializar los pesos.

Para inicializar los pesos con una distribución normal, podemos utilizar la función `numpy.random.normal`. Esta función permite especificar la media (`loc`), la desviación estándar (`scale`) y el número de valores a generar (`size`). Por ejemplo, `np.random.normal(loc=0, scale=1, size=(2,))` genera dos valores aleatorios con media 0 y desviación estándar 1. Esta inicialización es útil para romper la simetría y facilitar el aprendizaje del perceptrón.

```
# Crear una instancia del perceptron
classifier = Perceptron(n_entradas=2,f_activacion=sigmoid)
print("Pesos iniciales: ", classifier.w)
print("Bias inicial: ", classifier.b)

Pesos iniciales: [0. 0.]
Bias inicial: 0.0
```

```
# Inicializacion de parametros (1 punto)
classifier.w = np.random.normal(loc=0, scale=1, size=(2,))
classifier.b = np.random.normal(loc=0, scale=1)
print("Pesos despues de la inicializacion: ", classifier.w)
print("Bias despues de la inicializacion: ", classifier.b)

Pesos despues de la inicializacion: [1.5430634 0.6324931]
Bias despues de la inicializacion: -1.1371742479967
```

Función de pérdida

La función de pérdida nos indica que tanto nos estamos equivocando en el conjunto de datos completo.

En este notebook usaremos la suma de los errores absolutos

$$L = MAE = \sum_i^N |y^i - \hat{y}^i$$

```
# Implemente la funcion de perdida (1 punto)
def perdida(y_true, y_pred):
    # Inicializamos el error acumulado
    error = 0
    for i in range(len(y_true)):
        error += abs(y_true[i] - y_pred[i])

    return error

y_pred = classifier.inferencia(X)
print("Perdida inicial (IoS): ", perdida(y, y_pred))

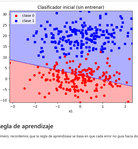
Perdida inicial (IoS): 79
```

A continuación visualizamos el rendimiento del clasificador de forma gráfica

```
x1, x2 = np.meshgrid(
    np.linspace(X_train[:,0].min(), X_train[:,0].max(), num=100),
    np.linspace(X_train[:,1].min(), X_train[:,1].max(), num=100)
)

grid = np.vstack([x1.reshape(), x2.reshape()])
y_pred = np.reshape(classifier.inferencia(grid), x1.shape)

fig = plt.figure()
ax = fig.subplots()
display = DecisionBoundaryDisplay(xx0=x1, xx1=x2, response=y_pred)
display.plot(ax=ax, plot_method='contour', cmap=cmap, alpha=0.3)
ax.ravel()[0].set_title('')
ax.set_ylabel('$x_2$')
ax.set_xlabel('$x_1$')
ax.scatter(X_train[y == 1], X_train[y == 1], c=cmap(1), label='Clase 0')
ax.scatter(X_train[y == 0], X_train[y == 0], c=cmap(0), label='Clase 1')
ax.set_title('Clasificador inicial (sin entrenar)')
ax.legend(loc='upper left')
plt.show()
```



Regla de aprendizaje

Primero, recordemos que la regla de aprendizaje se basa en que cada error no gua hacia donde mover los pesos. Es decir usamos el error residual

$$e^i = y^i - \hat{y}^i$$

Una vez definido el error residual observamos que la regla de aprendizaje se calcula como

$$W^{t+1} = W^t + \eta(y - \hat{y})X$$

Y se repite el cálculo tantas veces como deseemos.

```
# Implemente la regla de aprendizaje (2 puntos)
class Regla(BaseEstimator):
    def __init__(self, tasa_aprendizaje=0.1):
        self.tasa_aprendizaje = tasa_aprendizaje

    def actualizar(self, perceptron, X, y):
        for x_i, y_i in zip(X, y):
            output = perceptron.inferencia([x_i])
            error = target - output
            perceptron.b += self.tasa_aprendizaje * error # w2 Actualizar los pesos
            perceptron.w += perceptron.b * self.tasa_aprendizaje * error # w1 Actualizar el bias

# Configuracion del entrenamiento
tasa_aprendizaje = 0.5
n_epocas = 300 #Configura el numero de epocas

optimizer = Regla(tasa_aprendizaje=tasa_aprendizaje)

historial_perdida = []

for epoca in range(n_epocas):
    y_pred = classifier.inferencia(X)
    optimizer.actualizar(classifier, X, y)
    print("Epoca [{}/{}] Perdida (IoS): {}".format(epoca, n_epocas, perdida(y, y_pred)))
    historial_perdida.append(perdida(y, y_pred))
    optimizer.actualizar(classifier, X, y)

# Visualizar la perdida a lo largo de las epocas
fig, ax = plt.subplots(figsize = (5, 5)).historial_perdida, marker='o')
ax.ravel()[0].set_ylabel('Perdida (IoS)')
ax.ravel()[0].set_xlabel('Epoca')
ax.ravel()[0].set_title('Evolución de la pérdida durante el entrenamiento')
plt.show()
plt.show()

Epoca [0/300] Perdida (IoS): 77
Epoca [1/300] Perdida (IoS): 76
Epoca [2/300] Perdida (IoS): 75
Epoca [3/300] Perdida (IoS): 74
Epoca [4/300] Perdida (IoS): 73
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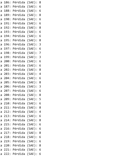


```
x1, x2 = np.meshgrid(
    np.linspace(X_train[:,0].min(), X_train[:,0].max(), num=100),
    np.linspace(X_train[:,1].min(), X_train[:,1].max(), num=100)
)

grid = np.vstack([x1.reshape(), x2.reshape()])
y_pred = np.reshape(classifier.inferencia(grid), x1.shape)

fig = plt.figure()
ax = fig.subplots()
display = DecisionBoundaryDisplay(xx0=x1, xx1=x2, response=y_pred)
display.plot(ax=ax, plot_method='contour', cmap=cmap, alpha=0.3)
ax.ravel()[0].set_title('')
ax.set_ylabel('$x_2$')
ax.set_xlabel('$x_1$')
ax.scatter(X_train[y == 1], X_train[y == 1], c=cmap(1), label='Clase 0')
ax.scatter(X_train[y == 0], X_train[y == 0], c=cmap(0), label='Clase 1')
ax.set_title('Clasificador despues del entrenamiento')
ax.legend(loc='upper left')
plt.show()
```